Geometric Modeling of Flagella in Single-Cell Organisms

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Introduction and Objectives

We add to the immersed boundary method discussed in Dr. Sookkyung Lim's paper, "Dynamics of a Closed Rod with Twist and Bend in Fluid." Namely, we

1. implement helical geometry to generalize flagellar dynamics to various relevant shapes.
2. develop explicit solutions for force and torque from the flagella to its ambient fluid for these shapes.

Kirchhoff Rod Theory and the Immersed Boundary (IB) Method

Kirchhoff Rod Theory

The Kirchhoff Rod Model is a mathematical framework designed to analyze the dynamics of filaments whose cross-sectional area is much smaller than their length. The model describes the rod as a three-dimensional space curve $\mathbf{X}(\mathbf{s})$ accompanied by an orthonormal triad $\{\mathbf{D}^1, \mathbf{D}^2, \mathbf{D}^3\}$ at each point that keeps track of the local orientation of the material. The theory of Kirchhoff for a thin elastic rod has been developed to find stable and unstable equilibrium configurations, and can be performed either in the absence or presence of a fluid. For our purposes, we analyze the dynamics of flagella in their ambient fluid. In the standard Kirchhoff rod model, one of the vectors in the material frame $\{\mathbf{D}^1, \mathbf{D}^2, \mathbf{D}^3\}$, namely $\mathbf{D^3}$, is constrained to be tangent to the space curve. Also, the Kirchhoff rod is constrained to inextensibility: the rod can bend and twist, but it cannot stretch. Because of the no-slip condition at the boundary of a viscous fluid, these constraints imply complicated boundary conditions in the fluid in which the rod is immersed. Certain conditions on the fluid velocity field need to be imposed along the length of the space curve, making simulations rather complicated.

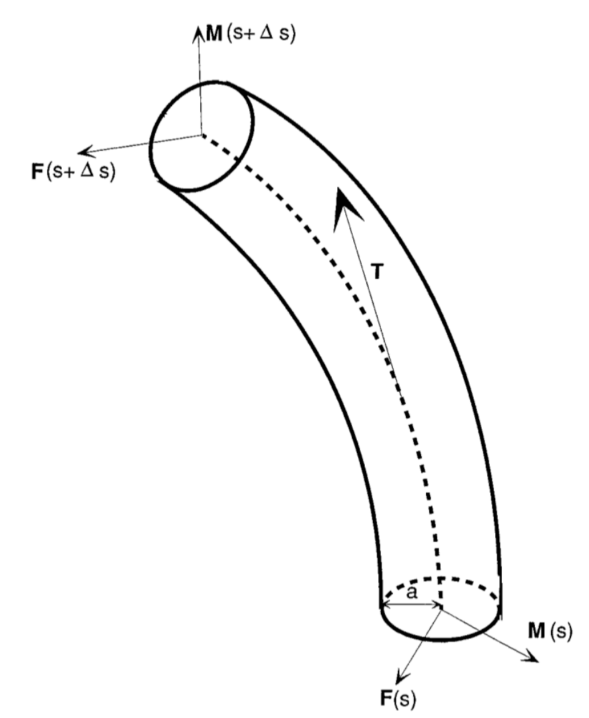


Figure 1. A visualization of the Kirchhoff Rod.

Immersed Boundary (IB) Method

The IB method avoids these complicated constraints on the fluid velocity by allowing

1. the rod to stretch, deviating slightly from the Kirchhoff rod theory of inextensibility.
2. the orientation of the triad $\{\mathbf{D}^1, \mathbf{D}^2, \mathbf{D}^3\}$ to deviate from alignment with the direction of the space curve, as opposed to constraining the triad to move in the direction of the centerline of the rod.

Instead of imposing the constraints of the Kirchhoff rod model exactly, Lim creates the following elastic energy function, which tends to maintain these constraints

$$E = \frac{1}{2}\int \left[a_1 \left(\frac{d\mathbf{D}^2}{ds}\cdot \mathbf{D}^3\right)^2 + a_2 \left(\frac{d\mathbf{D}^3}{ds}\cdot \mathbf{D}^1\right)^2 + a_3 \left(\frac{d\mathbf{D}^1}{ds}\cdot \mathbf{D}^2\right)^2 \right. \tag{1}$$

$$\left. + b_1 \left(\mathbf{D}^1 \cdot \frac{d\mathbf{X}}{ds}\right)^2 + b_2 \left(\mathbf{D}^2 \cdot \frac{d\mathbf{X}}{ds}\right)^2 + b_3 \left(\mathbf{D}^3 \cdot \frac{d\mathbf{X}}{ds} - 1\right)^2 \right] ds$$

where a_1, a_2, a_3, b_1, b_2 and b_3 are given material parameters of the rod. For example,

- $a_1 \left(\frac{d\mathbf{D}^2}{ds}\cdot \mathbf{D}^3\right)^2$ penalizes bending away from the $\mathbf{D^1}$ triad component;
- $b_3 \left(\mathbf{D}^3 \cdot \frac{d\mathbf{X}}{ds} - 1\right)^2$ penalizes the stretching of the rod.

Note that Lim's energy penalty function returns zero if the triad $\{\mathbf{D}^1, \mathbf{D}^2, \mathbf{D}^3\}$ is exactly oriented to the centerline of the rod, meaning in this instance there is no deviation from the Kirchhoff rod theory. Lim's energy penalty function is derived from a Lagrangian variational argument. It is shown that Lim's definitions of force and torque result in the energy penalty satisfying the Kirchhoff rod theory constraints. In particular, Lim derives the following force and moment balance equations that lay the foundation for analyzing the dynamics between the rod and the fluid

$$0 = \mathbf{f} + \frac{\partial \mathbf{F}}{\partial s}, \tag{2}$$

$$0 = \mathbf{n} + \frac{\partial \mathbf{N}}{\partial s} + \frac{\partial \mathbf{X}}{\partial s} \times \mathbf{F}, \tag{3}$$

where $\mathbf{F}$ and $\mathbf{N}$ are the force and torque from the rod, and $\mathbf{f}$ and $\mathbf{n}$ are the force and torque from the fluid. The forces and torques are defined as

$$\mathbf{F} = F^1\mathbf{D}^1 + F^2\mathbf{D}^2 + F^3\mathbf{D}^3, \tag{4}$$

$$\mathbf{N} = N^1\mathbf{D}^1 + N^2\mathbf{D}^2 + N^3\mathbf{D}^3, \tag{5}$$

$$\mathbf{f} = f^1\mathbf{D}^1 + f^2\mathbf{D}^2 + f^3\mathbf{D}^3, \tag{6}$$

$$\mathbf{n} = n^1\mathbf{D}^1 + n^2\mathbf{D}^2 + n^3\mathbf{D}^3, \tag{7}$$

where F^1, F^2, F^3 depend on the space curve $\mathbf{X}(s)$ and the triad $\{\mathbf{D}^1, \mathbf{D}^2, \mathbf{D}^3\}$, and N^1, N^2, N^3 depend on the triad.

Mathematical Formulation for an Equilibrium Configuration

We add to the IB method by developing explicit solutions for force and torque in an equilibrium configuration for new relevant flagellar shapes. Equilibrium between the rod and the fluid is achieved when $\mathbf{f}$ and $\mathbf{n}$ are both zero in Equations 2 and 3. The utility of finding explicit solutions includes

1. understanding the shapes and configurations a rod can take under its own elasticity,
2. validating numerical methods or simulations since recovering the known equilibrium configuration is a crucial benchmark condition,
3. providing a foundation for perturbation and dynamic analysis, as fluid-interacting models are first built around adding small deviations from equilibrium,
4. visualizing analytical solutions directly from the geometry of the rod.

In our relevant literature, Lim focuses on using the IB method for a closed circular rod. To find an equilibrium configuration for this circular rod in a horizontal plane, Lim introduces cylindrical coordinates (r, θ, z) with unit vectors $(r(\theta), \theta(\theta), z)$. We let $s = r_0\theta, 0 \leq s \leq 2\pi$ be the material coordinate of the rod along the circumference of the circle, with r_0 being the radius of the circular rod in its unstretched state. We now define the curve $\mathbf{X}(s)$ and the triad $\{\mathbf{D}^1, \mathbf{D}^2, \mathbf{D}^3\}$ in the cylindrical polar coordinates:

$$\mathbf{X}(s) = r_1\,\mathbf{r}(s/r_0),$$

$$\mathbf{r}(\theta) = (\cos\theta, \sin\theta, 0), \qquad \mathbf{D}^3(s) = (\cos\beta)\,\boldsymbol{\theta}(s/r_0) + (\sin\beta)\,\mathbf{z},$$

$$\boldsymbol{\theta}(\theta) = (-\sin\theta, \cos\theta, 0), \qquad \mathbf{E}(s) = -(\sin\beta)\,\boldsymbol{\theta}(s/r_0) + (\cos\beta)\,\mathbf{z},$$

$$\mathbf{z} = (0, 0, 1), \qquad \mathbf{D}^1(s) = \cos(ps/r_0)\,\mathbf{E}(s) + \sin(ps/r_0)\,\mathbf{r}(s/r_0),$$

$$\mathbf{D}^2(s) = -\sin(ps/r_0)\,\mathbf{E}(s) + \cos(ps/r_0)\,\mathbf{r}(s/r_0).$$

where p is a given integer number of twists in the rod. We endeavor to find r_1 to give an explicit solution for how the rod stretches and $\sin\beta$ to give the orientation of the triad along the curve. Note that the triad also depends on $\cos\beta$, but finding $\cos\beta$ is trivial if we are given $\sin\beta$. Notice that r_1 satisfies the first condition of the IB method and that $\sin\beta$ satisfies the second condition of the IB method.

The following visualization gives the triad along a circle in the xy-plane for arbitrary r_1 and $\sin\beta$ values.

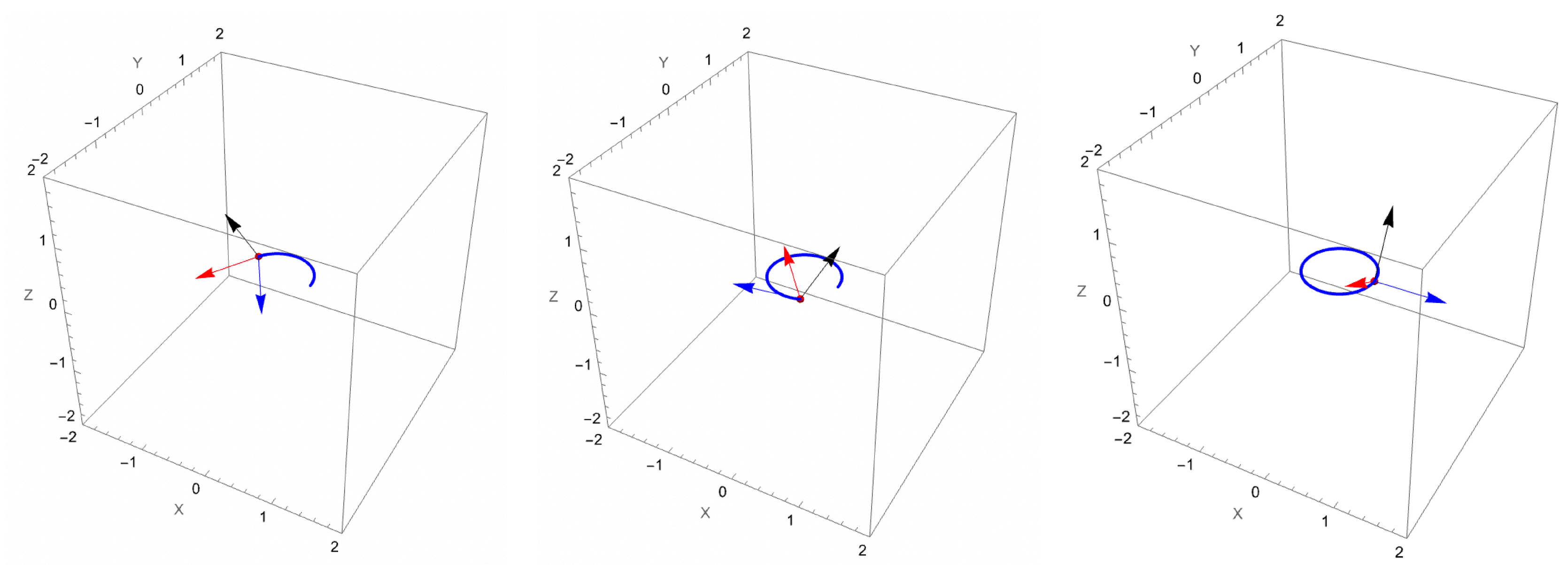


Figure 2. Triad representations at different positions s along the rod. The red arrow represents $\mathbf{D}^1$, the blue arrow represents $\mathbf{D}^2$, and the black arrow represents $\mathbf{D}^3$. Notice that $\mathbf{D^3}$ does not point in the direction tangent to the curve as per the original Kirchhoff rod constraints. Instead, the IB method imposes the energy penalty function that tends to maintain these constraints.

We now define the force and torque contributions from the rod with respect to the triad and the curve:

$$N^1 = a_1\frac{\partial \mathbf{D}^2}{\partial s}\cdot \mathbf{D}^3, \quad N^2 = a_2\frac{\partial \mathbf{D}^3}{\partial s}\cdot \mathbf{D}^1, \quad N^3 = a_3\frac{\partial \mathbf{D}^1}{\partial s}\cdot \mathbf{D}^2, \tag{8}$$

$$F^1 = b_1\mathbf{D}^1\cdot\frac{\partial \mathbf{X}}{\partial s}, \quad F^2 = b_2\mathbf{D}^2\cdot\frac{\partial \mathbf{X}}{\partial s}, \quad F^3 = b_3\left(\mathbf{D}^3\cdot\frac{\partial \mathbf{X}}{\partial s} - 1\right), \tag{9}$$

where the given material parameters a_1 and a_2 are the bending moduli of the rod about $\mathbf{D^1}$ and $\mathbf{D^2}$ and where a_3 is the twisting modulus of the rod. Note that in the context of Lim's paper, the rod is assumed to be isotropic, or $a_1 = a_2 \equiv a$ and $b_1 = b_2 = b_3 \equiv b$.

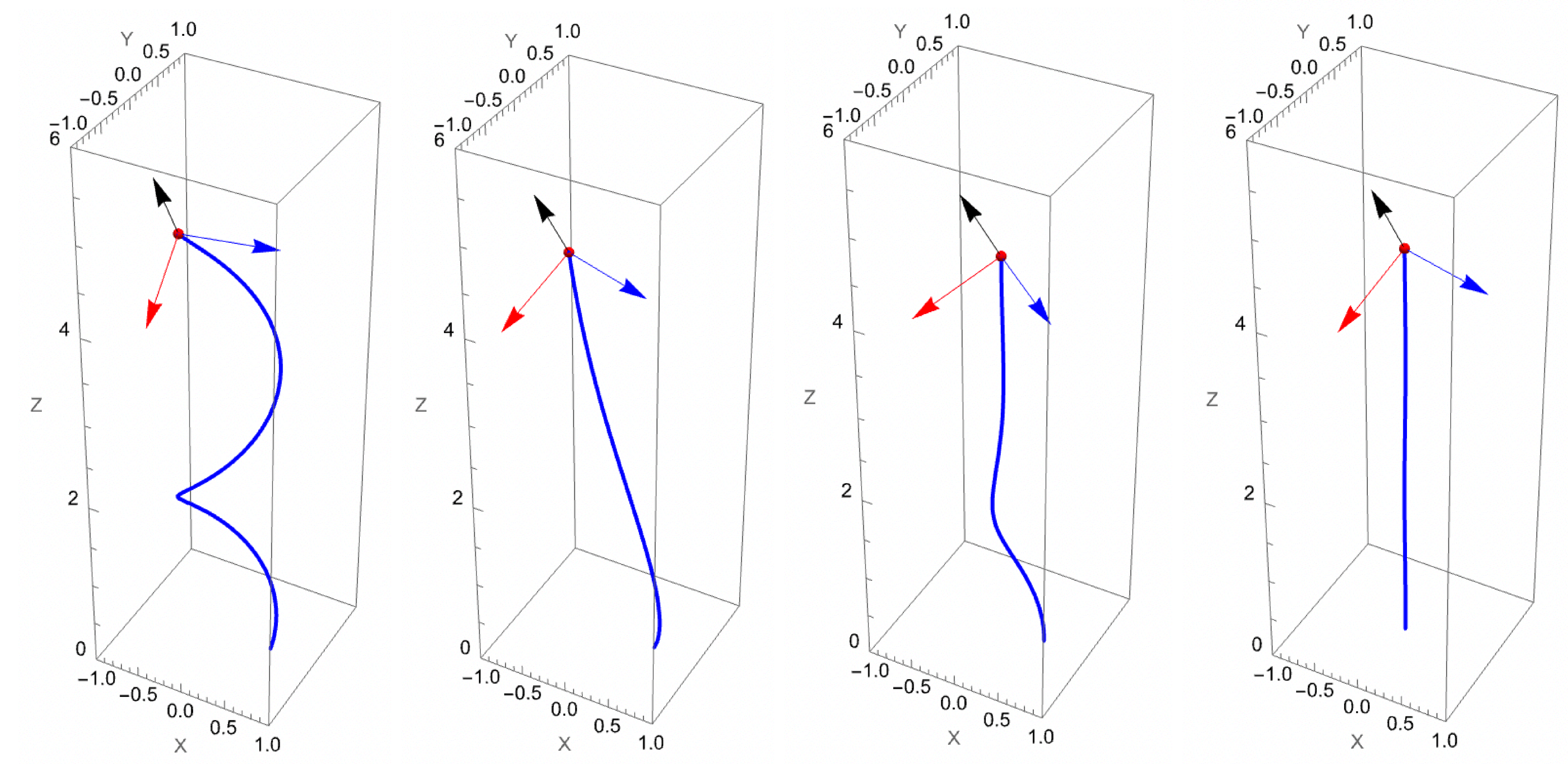
To solve the equilibrium configuration for r_1 and $\sin\beta$, we plug in Equations 8 and 9 into Equations 4 and 5. We then rewrite Equations 2 and 3 by plugging $\mathbf{f} = \mathbf{n} = 0$. Equations 10 and 11 give an example calculation for Equation 2, where r_1 is easily extractable from either the first or second component of the $\frac{\partial \mathbf{F}}{\partial s}$ vector. Similarly, $\sin\beta$ is extracted from Equation 3.

$$0 = \left\{ \frac{b\cos\left(\frac{s}{r_0}\right)(-r_1 + r_0\cos[\beta])}{r_0^2}, \quad \frac{b(-r_1 + r_0\cos[\beta])\sin\left(\frac{s}{r_0}\right)}{r_0^2}, \quad 0 \right\} \tag{10}$$

$$r_1 = r_0\cos\beta \tag{11}$$

Helical Geometry with Explicit Solutions for IB Method

We found parametrized solutions for forces and torques from the filament for relevant flagellar shapes. Our method for designing curves representing various flagellar shapes using Lim's framework was to use helices of variable pitch and radius. We noted that the equilibrium conditions of the circular filament in Lim's paper were valid when the circle lay in a horizontal plane, or when a sample unit vector pointing in the positive z-direction was orthonormal to the plane. We can evaluate forces and torques when the filament completes a parametrized circle when viewed from above, and using Lim's framework will not cause issues when solving for the equilibrium conditions. The four shapes were a helix with constant pitch and constant radius, a parabolic helix with constant pitch and constant radius, a helix of constant pitch and decreasing radius, and a helix of constant pitch much larger than its constant radius (an approximation for a straight line).



$\mathbf{X}(\mathbf{s}) = r_1\{\cos(\frac{s}{r_0}), \sin(\frac{s}{r_0}), \frac{s}{r_0}\}$

Helix: constant pitch, radius

$r_1 = r_0\cos\beta$

$\sin\beta = \frac{-a_3p + br_0^2\cos\beta}{-a + a_3 + br_0^2}$

$\mathbf{X}(\mathbf{s}) = r_1\{\cos(\frac{s}{r_0}), \sin(\frac{s}{r_0}), (\frac{s}{r_0})^2\}$

Parabolic helix: constant pitch

$r_1 = r_0\cos\beta$

$\sin\beta = \frac{-a_3p + 2br_0s\cos\beta}{-a + a_3 + br_0^2}$

$\mathbf{X}(\mathbf{s}) = r_1\{\cos(\frac{s}{r_0})e^{\frac{-s}{2}}, \sin(\frac{s}{r_0})e^{\frac{-s}{2}}, \frac{s}{r_0}\}$

Helix: decaying radius

$r_1 = \frac{-4e^{0.5s}r_0\cos\beta}{-4 + r_0^2 + 4r_0\tan\left(\frac{s}{r_0}\right)}$

$\sin\beta = \frac{e^{0.5s}(-a_3p + 0.01r_0^2\cos\beta)}{r_0^2(0.01 - 0.005r_0\tan(\frac{s}{r_0})) + e^{0.5s}(-0.01 + a_3)}$

$\mathbf{X}(\mathbf{s}) = r_1\{0.01\cos(\frac{s}{r_0}), 0.01\sin(\frac{s}{r_0}), \frac{s}{r_0}\}$

Helix with radius $\ll$ pitch

$r_1 = 100r_0\cos\beta$

$\sin\beta = \frac{-a_3p + 0.01r_0^2\cos\beta}{-0.01 + a_3 + 0.0001r_0^2}$

Project Requirements Table

#	Tollgate	Verification Method	Verified (Y/N)	Description
1	Understand problem, background research, and constraints	Define problem, identify relevant literature, & define methodology	Y	The Kirchhoff Rod is constrained to be inextensible and that one vector in the triad be tangent to the centerline. The IB method satisfied these constraints by implementing an energy penalty.
2	Brainstorm and choose direction	Analyze alternative or refined directions	Y	We wanted to find an explicit frame for any 3D curve, but decided against this for computational feasability.
3	Develop prototype solution	Make prototype changes, test, and optimize solution	Y	We iterated through solution methods such as solving a generalized set of complex coupled differential equations but decided on directly solving several simpler equations.
4	Test solution & validate	Evaluate prototype solution and validate final solution	Y	We developed a solution using helical geometry that satisfies the IB method constraints.

Future Work

1. Develop a solution that accounts for morphology in the filament by parametrizing the curve using different helical shapes. For example, flagella can have a helical shape, then suddenly become straight at a different part of the filament.
2. Find an explicit frame for any curve that solves the equilibrium configuration. One way to do this is to find a brute-force solution to a coupled set of differential equations involving the curve, the frame, and the first and second derivatives of the frame components.

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